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ORIGINAL RESEARCH

Physical and chemical crosslinked microcrystalline cellulose-polyvinyl alcohol hydrogel: freeze–thaw mediated synthesis, characterization and in vitro delivery of 5-fluorouracil

Sai Dileep Kumar Seera · Debashis Kundu · Tamal Banerjee

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Abstract The current work reports the synthesis of microcrystalline cellulose and polyvinyl alcohol co-polymerized crosslinked hydrogels both via physical and chemical crosslinking. While freeze–thawing ensured the physical crosslinking, the crosslinker, ethylene glycol diglycidyl ether ensured the chemical crosslinking. The obtained hydrogels were characterized by Fourier transform infrared spectroscopy, swelling ratio and rheology. The gels showed good equilibrium swelling ratio in deionized (DI) water and phosphate buffer saline (PBS) buffer. The morphological observation revealed porous and glossy nature of hydrogels. The synthesized hydrogels were used as a carrier for 5-Fluorouracil. The in vitro release of drug was performed in PBS buffer (pH = 7.4), leading to a cumulative release of 30–60% from all the chemically crosslinked gels. The kinetic release mechanism suggested the release of drug was mostly due to the controlled Fickian diffusion and swelling of gels.

Keywords Microcrystalline cellulose · Polyvinyl alcohol · Hydrogel · 5-Fluorouracil

Sai Dileep Kumar Seera and Debashis Kundu have equal contributions.

S. D. K. Seera · D. Kundu · T. Banerjee (✉)
Department of Chemical Engineering, Indian Institute of Technology Guwahati, Guwahati, India
e-mail: tamalb@iitg.ac.in

Introduction

The hydrogel is a soft polymeric network material and can be crosslinked physically or chemically to form three dimensional network of hydrophilic polymer that can accumulate large amount of water as well as biological fluids. The holding up of biological fluid as well as its structural similarity to that of cellular matrices, essentially makes the hydrogel a promising material for biomedical applications such as in drug delivery (Gupta et al. 2002), tissue engineering (Lee and Mooney 2001), wound dressing (Wang et al. 2007) and contact lens (Hyon et al. 1994). Polysaccharides are promising material as precursor of hydrogel due to the abundance of hydroxyl moieties, ease of crosslinking, abundant availability, biocompatibility and biodegradability. Apart from the natural polysaccharide, some synthetic polymers are also considered as precursor because of their biocompatibility and biodegradability.

Microcrystalline cellulose (MCC) is one of the forms of cellulose that can be synthesized by partially depolymerizing cellulose and thereby reducing the degree of polymerization. MCC holds advantage over conventional cellulose due to its high specific surface area (Mathew et al. 2005). Due to its biodegradability, MCC is widely used in pharmaceutical applications (Wang et al. 2018), in food industry as thickener, fat substitute, dairy product (Nsor-Atindana et al. 2017) and as stabilizer in creams (Adeyeye et al. 2002). Due

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to its mechanical strength, it is used as a reinforcing agent in the preparation of composites (Ni, Wen and Liu 2015). MCC has limited applications as a hydrogel material due to its insolubility in water and also been applied as an adsorbent for the remediation of heavy metal ions and dyes (El-Naggar et al. 2018). Therefore, for the dissolution, alternative solvents such as, sodium hydroxide/urea (Zhou and Zhang 2000), tetrabutyl ammonium fluoride/dimethyl sulfoxide (TBAF/DMSO) (Köhler and Heinze 2007) and ionic liquid (Liang et al. 2015) are explored. It can also be used as a drug delivery agent (Ting et al. 2014). Recently, MCC based hydrogels are used for the in vitro delivery of Cephalexin (Kundu and Banerjee 2020). Ye et al. (2017) reported fabrication of novel deformation induced anisotropic tough cellulose hydrogel with high stability as promising material in the field of tissue engineering. Further, the anisotropic cellulose hydrogels are also explored in cardiomyocytes (Ye et al. 2018). Additionally, the same research group reported single and dual crosslinked high strength cellulose hydrogels having attractive mechanical properties (Ye et al. 2019).

Polyvinyl alcohol (PVA) is one of the synthetic water soluble polymer having applications in the field of medicine, paper and food packaging industry (Tripathi et al. 2009) primarily due to its biocompatibility, nontoxicity and good mechanical strength. The presence of a pendant hydroxyl moiety makes it an excellent hydrophilic material thereby making crosslinking with other materials a feasible option. PVA hydrogels can be synthesized by various methods such as chemical crosslinking (Bo 1992), γ-irradiation (Park and Nho 2003), and physical crosslinking by freeze–thawing (Watase and Nishinari 1988). Among these methods, freeze–thaw has an advantage, because of its easy synthesis route with lesser number of intermediate steps. In the freeze–thaw process, upon freezing the PVA solution leads to the partial crystallization of PVA which enhances the formation of microcrystalline structure. Later, the super molecular structure is formed upon increasing freeze–thaw cycles (Stauffer and Peppas 1992). Physical crosslinking between synthetic polymers and polysaccharides may not always give the desired results. So to overcome this problem chemical crosslinkers are being used. Different forms of cellulose like bacterial cellulose (Wang et al. 2010), carboxymethyl cellulose (El Salmawi 2007), cellulose nano whiskers (Gonzalez et al. 2014) can be incorporated into PVA to make films or hydrogels which have potential applications in biomaterial (Wang et al. 2010), scaffolds for cell delivery (Drury and Mooney 2003), wound dressing (Gonzalez et al. 2014) and controlled drug release (Kim et al. 2014).

5-Fluorouracil (5-Fu), analogues to uracil holds a fluorine atom in the place of hydrogen at the C5 position of uracil. 5-Fu is widely used in the treatment of cancer like colorectal and aerodigestive tract, when combined with other chemotherapeutic agents and gave improvement in breast and brain cancer. However, 5-Fu has the greatest impact on colorectal cancer. It inhibits the release of thymidylate synthase (TS) by releasing its metabolites which are broken down by the liver and then by incorporating the essential metabolites into ribonucleic acid (RNA) and deoxyribonucleic acid (DNA) (Longley et al. 2003). The major clearance of 5-Fu is kidney and lung, and about 20% of 5-Fu is excreted as parent drug from the body (Pinedo and Peters 1988). Limited studies are being reported in the literature on the 5-Fu release from various hydrogels. Kan and Li reported the release of 5-Fu from chemically crosslinked methacrylic acid and 2-hydroxyethyl methacrylate hydrogels (Kan and Li 2013). Further Minhas et al. reported polyvinyl alcohol and polymethacrylic acid crosslinked hydrogels for the in vitro release of 5-Fu (Minhas et al. 2013).

In this study, we have synthesized a series of PVA-MCC based hydrogels by changing the molar feed ratio of PVA to MCC using different solvents such as dimethyl sulfoxide (DMSO) and NaOH/urea. Chemically crosslinked gels were prepared with the addition of ethylene glycol diglycidyl ether (EGDE) crosslinker. The crosslinking happens between the hydroxyl moieties of MCC and PVA, with the opening of the epoxide ring present in EGDE, in the highly basic medium (Kundu et al. 2019). Fourier transform infrared (FTIR) spectroscopy was used for structural characterization whereas morphological observation was performed by optical microscope and field emission scanning electron microscope (FESEM). Rheological properties of as-synthesized gels such as flow measurement, amplitude sweep, and frequency sweep were studied at 25 °C. Physical properties of the synthesized hydrogels were determined by the swelling ratio (*SR*). The synthesized gels were used for the encapsulation of 5-Fu and subsequent in vitro release in phosphate buffer saline (PBS) of pH = 7.4.

Materials and methods

Materials

PVA (86–89% hydrolyzed) was purchased from Alfa-Aesar and MCC was purchased from Merck. 5-Fu and EGDE were purchased from TCI India. Auxiliary chemicals such as urea, sodium hydroxide and hydrochloric acid (37 wt %) were purchased from Merck while potassium bromide was purchased from Sigma-Aldrich. All the above chemicals were used as received. Deionized (DI) water was used as prepared from the in-house Millipore water synthesis unit (make: Millipore, model: ELix-3).

Preparation of hydrogels

NaOH/urea as a solvent

The synthesis of chemically crosslinked MCC-PVA hydrogel, given in Fig. 1, consists of three different steps, namely (i) preparation of precursors by freeze–thaw cycle, (ii) synthesis of hydrogel with the addition of crosslinker and (iii) washing and neutralization of gels. Initially, PVA precursor was prepared by dissolving PVA in 20 mL of DI water at 90 °C for 1 h under magnetic stirring. After complete dissolution of PVA, a calculated amount of NaOH and urea were added as reported in the literature (Zhou and Zhang 2000). Thereafter, MCC was added in the mixture and three such solutions were prepared by changing PVA to MCC in the feed mole ratio of 3:1, 5:1, 7:1. However, with the addition of NaOH/urea and MCC into PVA solution, the mixture did not turn into a homogeneous solution. After the first freeze–thaw step, the MCC and PVA solution has a thin layer of gel, which might be physically crosslinked gel and colloid of MCC. Upon stirring for 3 h, the formed layer of gel is disturbed and disappeared. This mixture was subjected to another 3 freeze–thaw cycles consisting 8 h of freezing at (− 20 °C) followed by 2 h of thawing at room temperature. After four cycles of freeze–thawing, MCC was completely dissolved and the physically crosslinked gels were obtained.

For chemically crosslinked gels, after the freeze–thaw step, as shown in step 2 of Fig. 1, EGDE was added dropwise into the solution by maintaining its temperature at 70 °C and continuously stirring the solution until gelation was observed. Next, the formed gels were cooled to room temperature and washed thoroughly with DI water to remove the excess amount of precursors. The obtained gels were subjected to

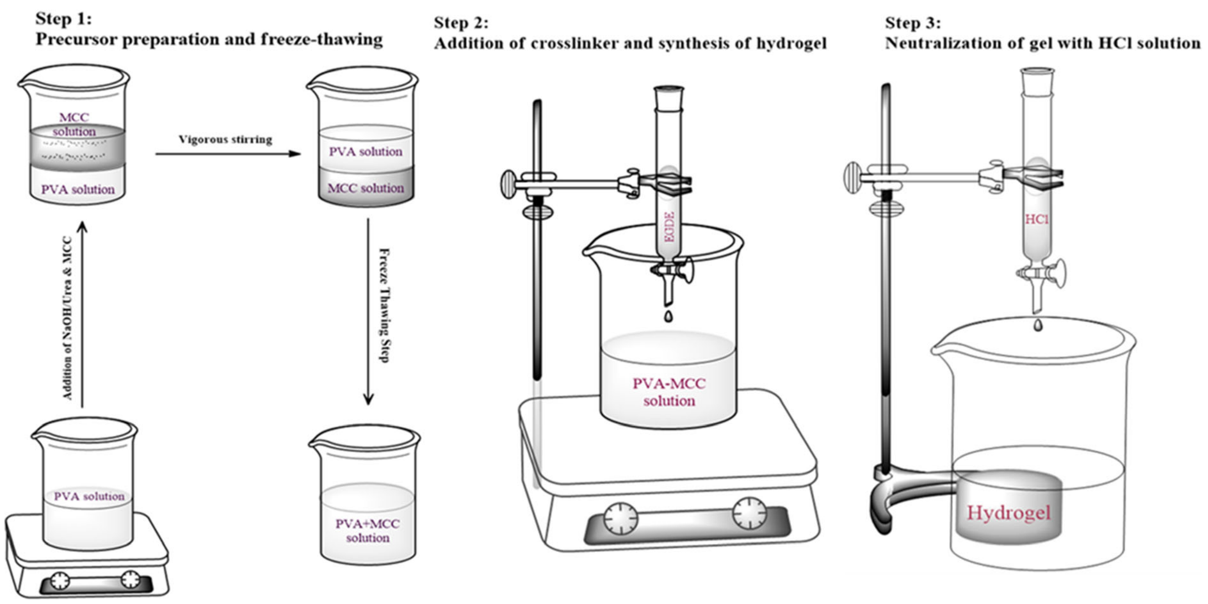


Fig. 1 Representation of chemically cross-linked hydrogel synthesis

neutralization using 0.1 M HCl solution and later lyophilized (make: Martin Christ, model: Alpha 2–4 LD). Schematic of the chemically crosslinked hydrogel using NaOH/urea as a solvent is shown in Fig. 2. Chemically crosslinked hydrogels using NaOH/urea as a solvent are referred to P3M1, P5M1, P7M1 according to mole ratio of PVA to MCC.

DMSO, NaOH/urea as a solvent

Preparation of hydrogels using DMSO, NaOH/urea as a solvent was similar to that of gel prepared by NaOH/urea as a solvent. Initially, MCC solution was prepared by dissolving in NaOH/urea solution. Two freeze–thaw cycles were performed consisting of 16 h of freezing and 2 h of thawing at room temperature to dissolve MCC. Separately, PVA was dissolved in DMSO at 40 °C under constant stirring. Later, different amounts of PVA solution (i.e. PVA to MCC in the mole ratio of 3:1, 5:1, 7:1) were slowly added to the MCC solution where gelation was observed within 4–5 min at room temperature (25 °C). Therefore, by using DMSO, NaOH/urea as solvents, we have obtained second set of physically crosslinked hydrogel. However, the DMSO containing solvent medium did not facilitate chemical crosslinking involving EGDE. Hence, no chemically crosslinked hydrogels were prepared with this route. The physically formed crosslinking gels were then washed with DI water to remove any unreacted monomers and solvent and stored for further applications. Physically crosslinked DMSO gels were referred to P3M1_D_Phy, P5M1_D_Phy, P7M1_D_Phy according to mole ratio of 3:1, 5:1, 7:1 of PVA to MCC.

Characterization

Rheology

Rheological investigations of as-synthesized hydrogels were performed on a rheometer (make: Anton Parr (Austria), model: Physica MCR 301) at 25 °C equipped with a cone-plate geometry (with a diameter of 50 mm and angle 1° with 0.1 mm gap between the plates). Flow behavior tests were carried out within the shear rate of 0.01–1000 s^{-1}. To find linear viscoelastic range (LVR), amplitude sweep test was carried in between the strain (%) ranging from 0.1 to 1000 at a constant frequency of 1 Hz. Oscillatory frequency tests were done in the frequency range of 0.01–100 Hz at a constant strain of 0.5% found from LVR. Loss tangent (tan δ), defined as the ratio between viscous to elastic response, was calculated by Eq. 1.

$$\tan\delta = \frac{G''}{G'} \quad (1)$$

Here, storage moduli is denoted by G' and loss moduli is given by G''.

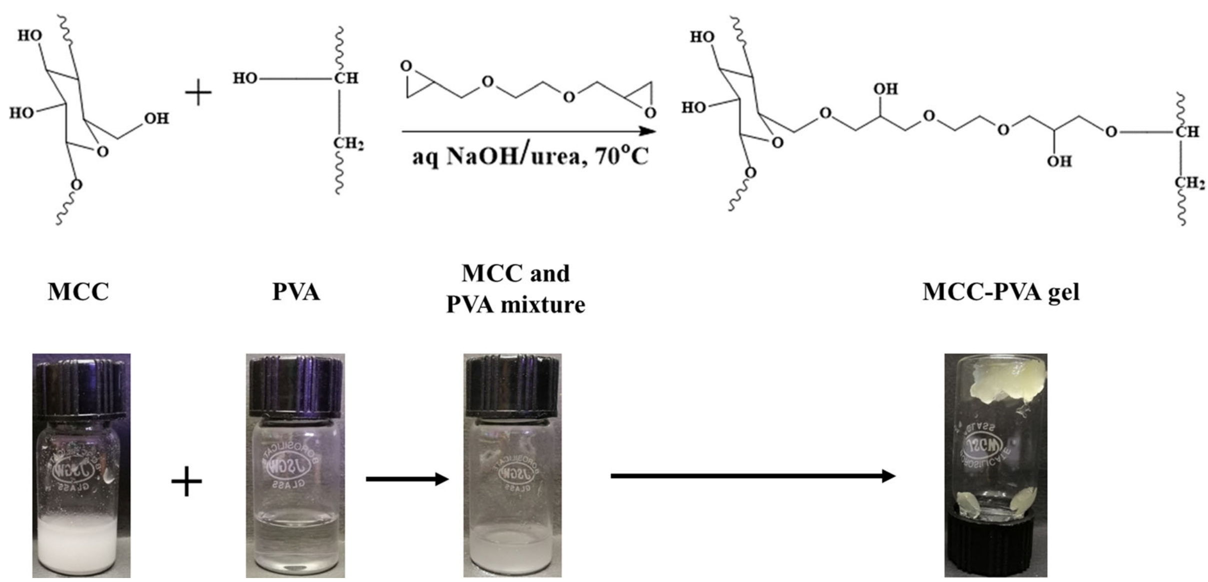


Fig. 2 Reaction scheme of the chemically crosslinked hydrogel in NaOH/urea solvent

Morphology

Freeze-dried hydrogels were used for observing the morphology in optical microscope and FESEM. For optical microscope (make: Carl Zeiss, model: Scope A1, AXIO, software: ZEN 2.3) the freeze-dried gel was deposited on a glass slide and covered with a cover glass. Further drug loaded hydrogel was observed with the same microscope in the fluorescence mode at 385 nm wavelength light. For FESEM (make: Carl Zeiss, model: Gemini 300), the freeze-dried gel was tip casted on a carbon tape which was placed on a stub and sputter-coated with a double layer of gold. The observations were carried at an optical zoom range of 1 µm and 200 nm with a potential of 3 kV.

FT-IR spectroscopy

Freeze-dried hydrogel sample was blended with an excess amount of dried Potassium bromide (KBr) into a fine mixture, and was pressed by a hydraulic press to form transparent pellets. These pellets were used for FTIR study (make: Shimadzu, model: IR Affinity-1) with a resolution of 4 cm^{-1} within 400–4000 cm^{-1} range using 30 scans for each sample.

Equilibrium swelling ratio

Swelling behavior of MCC-PVA hydrogels was determined by allowing pre-weighted freeze-dried hydrogel immersing in DI water and PBS solution for 48 h at 25 °C to reach equilibrium swelling condition. After 48 h of swelling, the gels were filtered using filter paper. The equilibrium weight of rehydrated gel was measured, and the swelling ratio of the gel was calculated by using the equation as given below.

$$Swelling\ Ratio(\%) = \frac{(W_{eq} - W_o)}{W_o} \times 100\% \tag{2}$$

where W_{eq} represents equilibrium weight of swelled gel and W_o is the initial weight of freeze-dried gel.

Drug loading efficiency

The 5-Fu drug solution was prepared by dissolving calculated amount of 5-Fu in DI water (0.5 mg mL^{-1}). The freeze-dried and pre-weighted hydrogel was immersed in 10 mL of 5-Fu solution and kept in an orbital shaker for 36 h at 25 °C and 120 rpm in dark condition. After 36 h, the drug-adsorbed gel was filtered and freeze-dried for subsequent in vitro release study. The obtained filtrate was then diluted appropriately to measure the absorbance by using UV–Vis spectrophotometer. The amount of 5-Fu loaded to the hydrogel carrier is determined from the following equation

$$\text{Drug Loading Efficiency (\%)} = \frac{W_t - W_r}{W_t} \times 100 \tag{3}$$

where W_t is the total amount of drug in the solution initially and, W_r is the amount of drug retained in the solution.

In vitro release of 5-Fu

The in vitro release of drug from the drug loaded hydrogels was performed in phosphate buffer saline (PBS, pH = 7.4). 5-Fu loaded freeze-dried hydrogel was immersed in 50 mL of PBS solution inside a conical flask. The conical flask was then placed in the same incubator at 37 °C. After 1 h of a time interval, 10 mL of buffer solution was collected and to maintain equal volume, 10 mL of fresh buffer solution was added to the flask. The method of sample collection was duly followed from literature (Kundu and Banerjee 2019). The collected buffer was filtered, appropriately diluted and the concentration of released drug was measured through UV–Vis spectrophotometer.

The in vitro 5-Fu release profile was fitted to Peppas, Higuchi, Zero-order, First-order models to find the kinetic mechanism of 5-Fu release. The kinetic models are given below:

$$\text{Peppas Model: } \frac{M_t}{M_\infty} = kt^n \tag{4}$$

$$\text{Higuchi Model: } \frac{M_t}{M_\infty} = k\sqrt{t} \tag{5}$$

$$\text{Zero-order Model: } \frac{M_t}{M_\infty} = kt \tag{6}$$

$$\text{First-order Model: } \frac{M_t}{M_\infty} = 1 - e^{-kt} \tag{7}$$

where M_t is the amount of drug (5-Fu) released in mg at time t and M_∞ is the total amount of drug loaded in the hydrogel thus $\frac{M_t}{M_\infty}$ is the fraction of drug released at time *t,* and k is constant which incorporates a characteristic of the drug and macromolecular network, n is diffusional exponent indicating kinetic release mechanism.

UV–Vis spectroscopy

Standard calibration curve of a drug (5-Fu) was prepared by plotting the known concentration of 5-Fu against absorbance. The known concentrations were prepared by diluting 0.5 mg mL^{-1} 5-Fu stock solution accordingly. Different calibration curves were plotted for DI water and PBS. Thereafter, the absorbance of the filtrate was measured with UV–Vis spectrophotometer (make: Shimadzu, model: UV-2600). The concentration of aliquots was calculated from the absorbance data plotted in the calibration chart.

Results and discussions

Rheological analysis

The viscosity of as-synthesized hydrogels is shown in Fig. 3 and performed at 25 °C by varying shear rate between 0.01 and 1000 s^{-1}. From the flow measurements, the hydrogels show shear thinning behaviour i.e. with increasing shear rate viscosity decreases. The reason attributes to the loosening of crosslinking structure between MCC and PVA with the increase in shear rate. Therefore, the transformation allows less resistance of PVA and MCC repeat units to deform and hence by increasing the shear rate, the decrease in viscosity is observed. Therefore, the shear-thinning behaviour of hydrogels are confirmed by fitting within a power-law model, apparent viscosity in terms of power-law is given as

$$\eta = m(\dot{\gamma})^{n-1} \tag{8}$$

where $\dot{\gamma}$ is the shear rate, η is the apparent viscosity, m is the consistency index, n is the power-law index. Table 1 represents the model fitted parameters for the hydrogels. For all the gels, the obtained power law index is less than unity which is a confirmation of shear-thinning behaviour with increasing shear rate.

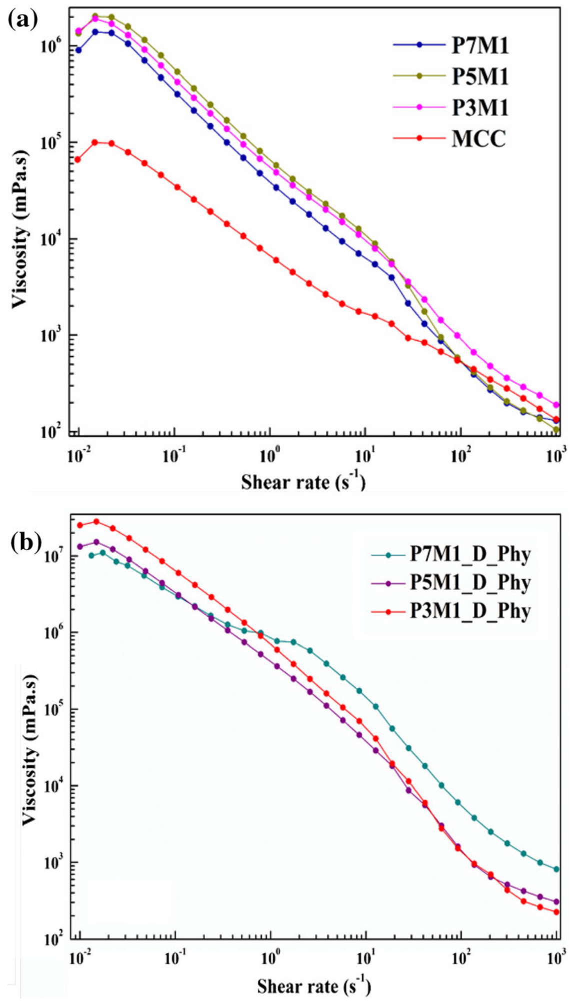


Fig. 3 Viscosity of hydrogels: **a** chemically crosslinked gels using NaOH/urea as solvent, **b** physically crosslinked gels using DMSO, NaOH/urea as solvent

Table 1 Power law parameters

Hydrogel	*m*	*n*	R^2
P7M1	3.94×10^4	0.118	0.993
P5M1	5.95×10^4	0.071	0.989
P3M1	5.75×10^4	0.144	0.995
MCC	0.76×10^4	0.404	0.991
P7M1_D_Phy	5.04×10^5	0.114	0.97
P5M1_D_Phy	2.92×10^5	0.038	0.989
P3M1_D_Phy	4.47×10^5	0.128	0.987

Further, we observe the viscosity of physically crosslinked gels in DMSO, NaOH/urea solvent, is higher by order of ten than chemically crosslinked gels in

NaOH/urea solvent. Lower viscous nature is observed in chemically crosslinked pure MCC gel in NaOH/urea solvent.

Amplitude sweep test is performed at 25 °C by varying the strain between 0.1 and 1000%. Domination of storage modulus over loss modulus is observed for all the gels until cross over point and shown in Fig. 4a and b. After cross over point, loss modulus dominates over storage modulus indicating breakage of the crosslinking structure of the gels. The storage modulus of physically crosslinked gels in DMSO, NaOH/urea solvent is 100–250 times higher than chemically crosslinked gels in NaOH/urea solvent. The higher storage modulus of physically crosslinked gels in DMSO, NaOH/urea is may be due to the formation of lumps by the macromolecules. It is found that the LVR from the amplitude sweep test is 0.5% of strain.

Further, the frequency sweep test is performed on all hydrogels at 25 °C, with a 0.5% strain by varying frequency from 0.1 to 100 Hz. There is no sign of crossover point from Fig. 4c and d within the frequency range. This is an indication of the phase transformation of gels which is not occurring at a given strain value of 0.5%. However, the domination of storage modulus over loss modulus is observed within the frequency range. For the physically crosslinked gels in DMSO with NaOH/urea, storage modulus is 15–50 times higher than the chemically crosslinked gels in NaOH/urea. From both amplitude and frequency sweep experiments, the storage modulus of physically crosslinked gels in DMSO or NaOH/urea is higher than chemically crosslinked gels in NaOH/urea. As stated earlier the higher storage modulus may result from the formation of lumps due to more crosslinking between MCC and PVA. Moreover on immersion of the physically crosslinked gels in water, there is no sign of swelling or absorption of water, a primary characteristic of a hydrogel. It may be due to the hindrance of crosslinking between precursors. After that, upon ageing degradation is observed in the case of physically crosslinked gels in DMSO and NaOH/urea. Thus from rheology results, further physical characterizations are not performed on physically crosslinked gels in DMSO, NaOH/urea.

Finally, loss tangent is calculated from Eq. 1, and Fig. 5 shows the corresponding plot of loss tangent vs frequency of both chemically crosslinked gels in NaOH/urea and physically crosslinked gels in DMSO and NaOH/urea. For all the hydrogels, loss tangent is less than 0.4 and no sign of phase change within the frequency range, indicating the gels are predominantly elastic.

FT-IR spectra

Figure 6 shows the FT-IR characterization of pure MCC, PVA and MCC-PVA hydrogel. Figure 6a represents the FT-IR characterization of pure MCC where the broad peak at 3345 cm^{-1} represents the stretching of O–H moiety and the other major peaks at 2901 cm^{-1}, 1645 cm^{-1}, 1431 cm^{-1}, 1163 cm^{-1}, 895 cm^{-1} corresponds to the stretching of C–H, stretching of C–O, asymmetric vibration of $-CH_2$ i.e. crystallinity band (El-Naggar et al. 2018), stretching of C–O–C and finally vibration of –CH corresponds to β-(1,4)-glycosidic linkage (Trache et al. 2016) respectively. Besides, the peak at 1032 cm^{-1} is assigned to C–O–C stretching of the ether linkage (El-Naggar et al. 2018). Figure 6b shows the FTIR spectra of pure PVA and the characteristic peaks are related to the hydroxyl and acetate moieties. The characteristic band at 3306 cm^{-1} links to the stretching of O–H moiety. The peak at 2940 cm^{-1} corresponds to the stretching of C–H from alkyl moiety. Because of the unreacted acetate moiety present in PVA, the peak at 1732 cm^{-1} is linked to the stretching of C=O and C–O of acetate moiety (Mansur et al. 2008). The crystallinity band of PVA can be observed at 1141 cm^{-1}, which is assigned to symmetric stretching of C–C or C–O stretching (Mansur et al. 2008). Figure 6c illustrates the FTIR spectra of PVA-MCC gel. The broad peak at 3421 cm^{-1} is O–H moiety stretching, and the dual peak at 2915–2882 cm^{-1} is because of C–H stretching. Because of hygroscopic nature of the gels, the bending of H–O–H of absorbed water is depicted at 1655 cm^{-1} (Kundu et al. 2019). The peak at 1458 cm^{-1} is assigned to the vibration of $-CH_2$, and the peak observed at 1076 cm^{-1} can be assigned to the bending of C–C. Interestingly the disappearance of two peaks at 890–897 cm^{-1} and 1160–1165 cm^{-1} in PVA-MCC hydrogel, typical of MCC with a characteristic of β-(1,4)-glycosidic linkage, are an indication that the crosslinking indeed took place between PVA and MCC with EGDE (Chang et al. 2008).

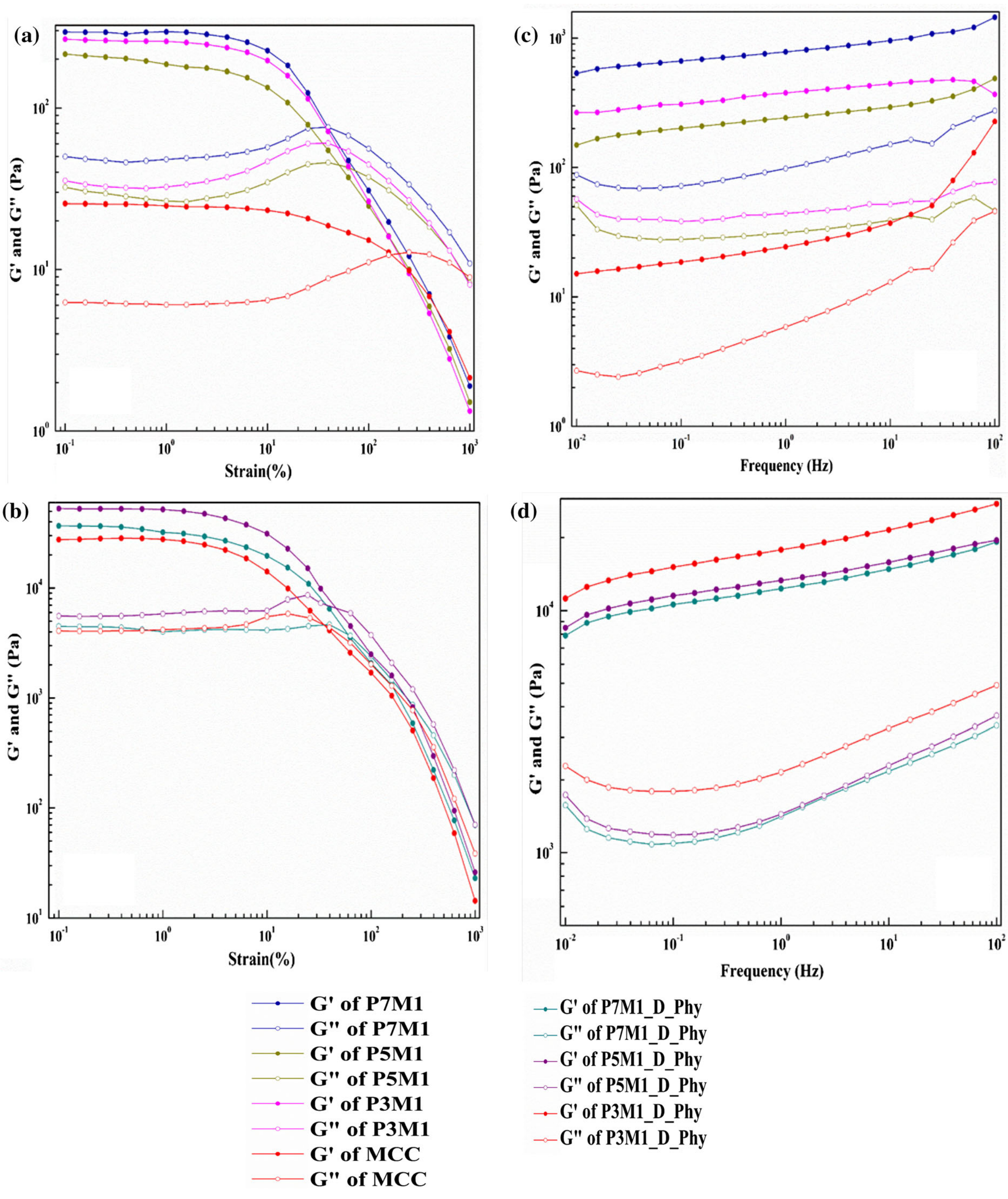


Fig. 4 Rheology of hydrogels: **a** amplitude sweep of chemically crosslinked hydrogels using NaOH/urea as solvent, **b** amplitude sweep of physically crosslinked hydrogels using DMSO, NaOH/urea as solvent, **c** frequency sweep of chemically crosslinked hydrogels using NaOH/urea as solvent and **d** frequency sweep of physically crosslinked hydrogels using DMSO, NaOH/urea as solvent

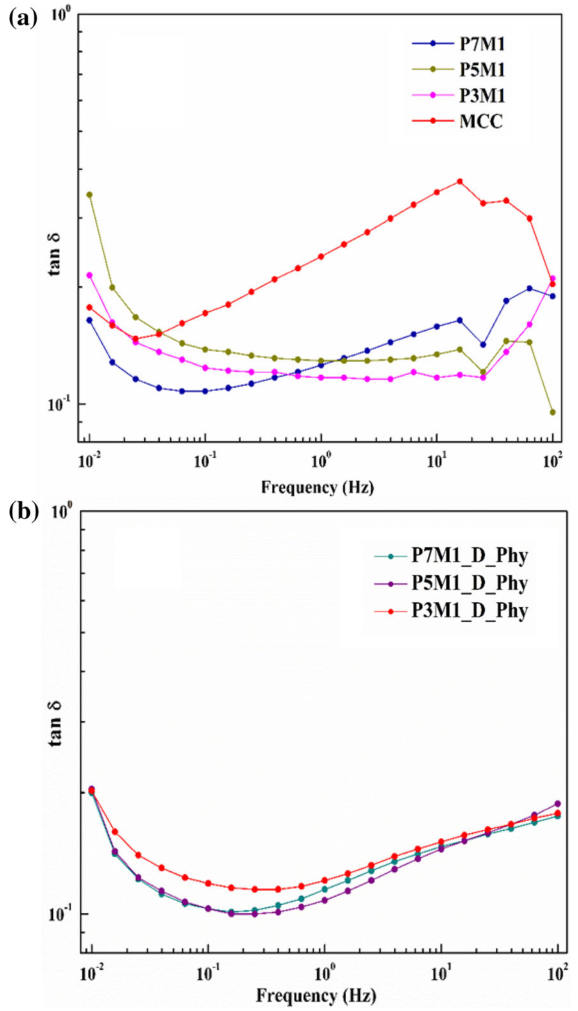


Fig. 5 Loss tangent of hydrogels: **a** chemically crosslinked gels using NaOH/urea as solvent, **b** loss tangent of physically crosslinked gels using DMSO, NaOH/urea as solvent

Morphology

Optical and FESEM micrographs of the freeze-dried hydrogel are illustrated in Figs. 7 and 8 respectively. The optical image of a hydrogel as shown in Fig. 7a, reveals the open macroporous structure. The gel appears to be shiny and glossy because of the presence of moisture in the pores. Figure 7b represents the 5-Fu loaded hydrogel viewed in the fluorescence microscope. The bright blue spots in the image correspond to the concentration of entrapped drug molecules where the drug loading cannot be seen all over the gel. The porous nature of the hydrogel can be observed in the FESEM image of Fig. 8. The random measurement of diameters of pores which are of different sizes of 670 nm and 1.042 µm as shown in Fig. 8b reveals the presence of large pores in the hydrogel matrix. These pores increase the availability of binding sites where the drug or water molecules can be encapsulated.

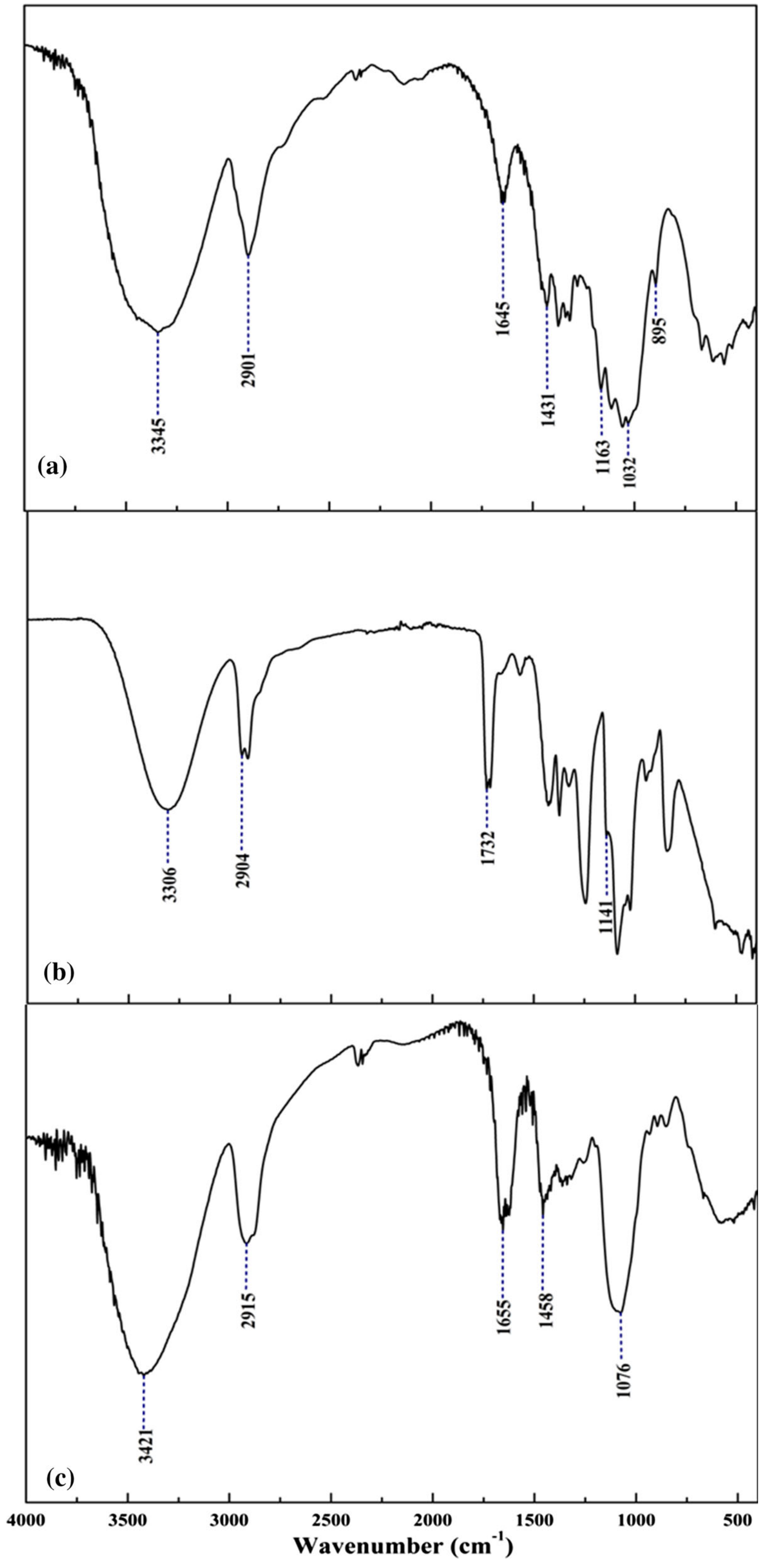
Swelling ratio

Figure 9 shows the swelling ratio (*SR*) of MCC-PVA hydrogels measured in DI water and PBS buffer. Freeze-dried and swollen hydrogel images are shown in Fig. 9a and b respectively. Amidst all the gels, P5M1 is having the highest swelling ratio of around 1404.24% and P7M1 bearing the lowest value of about 1136.75%. The swelling ratio of P3M1 is around 1336.9% which is in between P5M1 and P7M1 gels. From the amplitude and frequency sweep experiment results, P7M1 has higher storage modulus among three gels which is an indication of a higher crosslinking between MCC and PVA. Because of more crosslinking in P7M1 gels, hindrance occurs in the hydrogel matrix, which in turn reduces the swelling ratio. As a result, a low swelling ratio can be observed for P7M1 gel from the *SR* measurement. For P5M1 gel, contrary to P7M1, is having more swelling ratio and less crosslinking between MCC and PVA. Amplitude, frequency sweep results show the low storage modulus, an indication of less crosslinking between precursors. The P3M1 gel lies in between the P5M1 and P7M1 gels in case of both swelling ratio and storage modulus. Therefore, we conclude that the optimum PVA and MCC molar ratio is 5:1 to have higher swelling results. The order of swelling ratio is P5M1 > P3M1 > P7M1, and interestingly the order of storage modulus is P7M1 > P5M1 > P3M1 showing less crosslinking gels have more swelling capacity and low storage modulus. In other words, higher the crosslinking, lower the swelling and higher the storage modulus. The trend of *SR* in PBS buffer is the same as in DI water, i.e., P5M1 > P3M1 > P7M1 but having more *SR* than in DI water. The trend of *SR* for hydrogels is in PBS > DI water.

Drug loading and in vitro release of 5-Fu

The drug (5-Fu) loading in MCC-PVA chemically crosslinked hydrogels is recorded in between 89 and 98%. P5M1 gel possesses the highest value of 97.19%

Fig. 6 FT-IR spectra: **a** pure MCC, **b** pure PVA, **c** MCC-PVA crosslinked gel



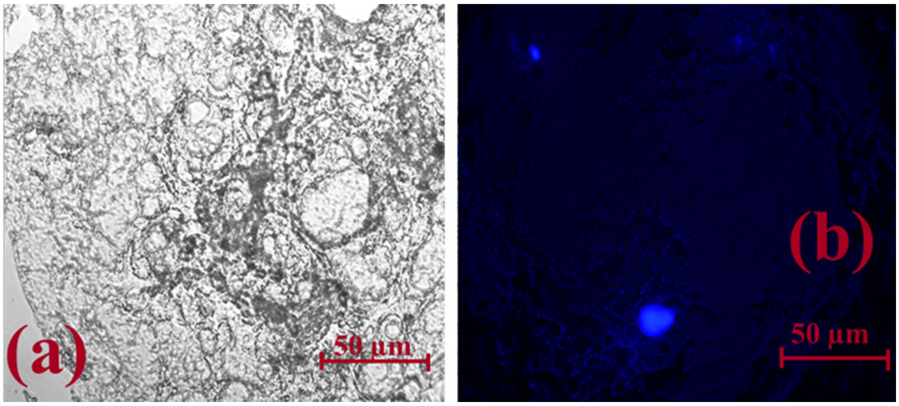


Fig. 7 Optical microscopic images of hydrogel: **a** MCC-PVA gel and **b** 5-Fu loaded gel

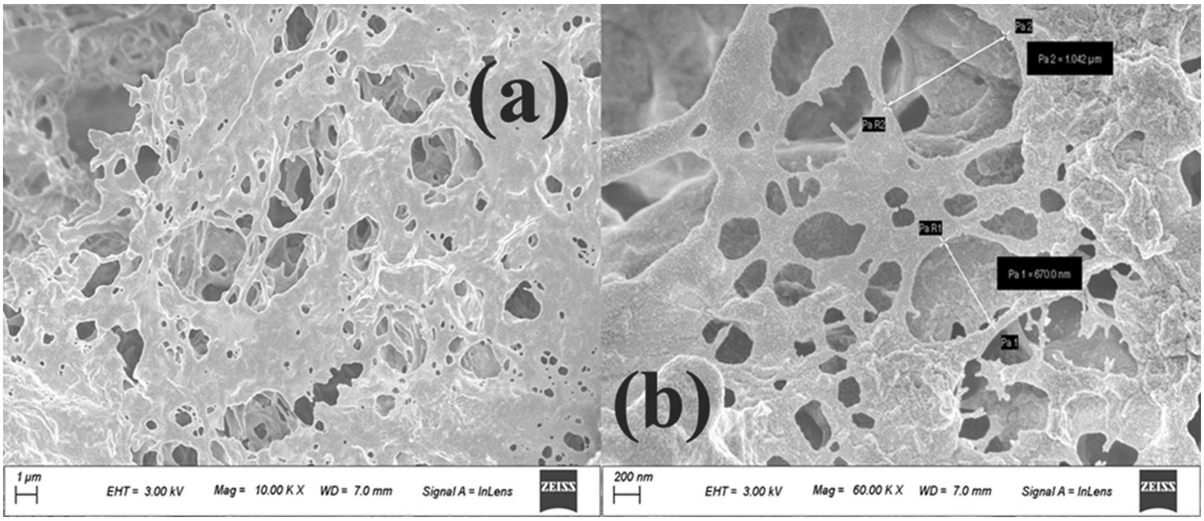


Fig. 8 FESEM micrographs of hydrogel: **a** MCC-PVA gel at 1 µm and **b** MCC-PVA gel at 200 nm

followed by P3M1 with 93.93% and P7M1 with 89.66%. Loading of drug follows a similar trend of the swelling ratio and opposite to storage modulus (i.e., the crosslinking) which can be inferred that due to the hindrance of crosslinking, the drug loading is decreasing. Further, the increment of drug loading can be observed with the increasing trend of swelling ratio.

The in vitro release of 5-Fu from the hydrogel in PBS buffer is shown in Fig. 10. The cumulative release of 5-Fu ranges from 30 to 60% in the crosslinked hydrogels. The highest release is seen in P5M1 with 59.35%, followed by P7M1 with 38.7% and P3M1 with 31.56%. Based on the results of drug loading efficiency and cumulative release, it can be assumed that the size of the drug molecule, the pore size of hydrogel and crosslinking between MCC and PVA can influence the release of drug from the hydrogel. The porous nature of the hydrogel is proved from the FESEM images where the diameter of couple of pores is shown. The pore diameter is bigger than the size of the 5-Fu. Therefore, the drugs can be loaded and subsequently released from the pores of the hydrogel. Further, as the pores are capable of entrapping the drug molecule, the release of 5-Fu from the hydrogel is restrained leading to the lower release from the gels. However, the release of 5-Fu may be hindered by the crosslinking between MCC and PVA since less hindrance is observed in P5M1 than P7M1, P3M1. This leads to a higher release of drug from P5M1 hydrogel.

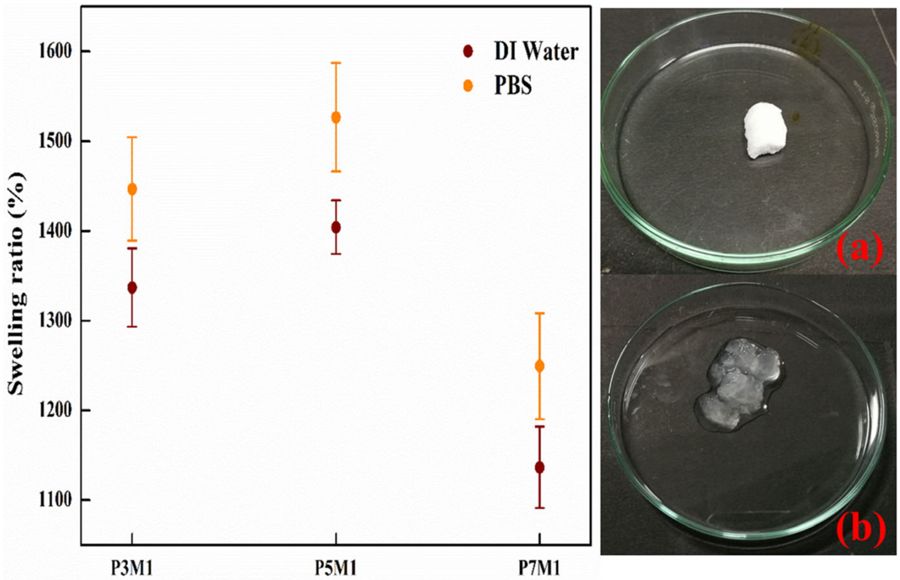


Fig. 9 Swelling ratio of hydrogels in DI water and PBS, **a** freeze-dried hydrogel, **b** swollen hydrogel in DI water

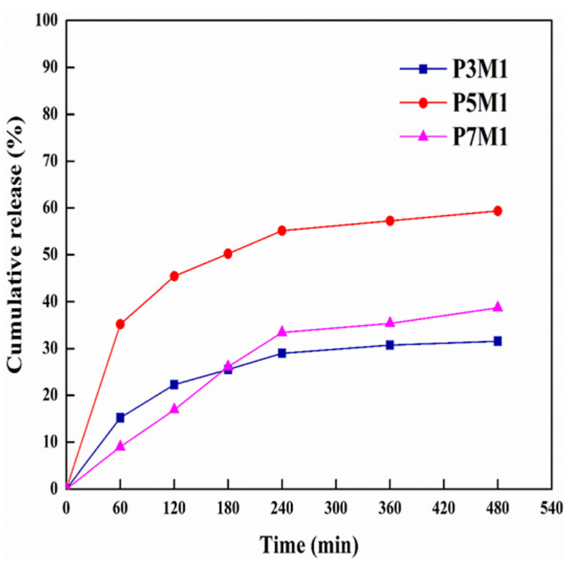


Fig. 10 In vitro release of 5-Fu in PBS buffer

The release of 5-Fu behavior can be explained in two stages, i.e. burst release (stage 1) and sustained or gradual release (stage 2). For the release mechanism of gels, the kinetic equations of Peppas, Higuchi, Zero-order, First-order models are used to fit the release data. The diffusion exponent and correlation coefficient obtained in each model are tabulated in Table 2. The Peppas model fits most appropriately for the release data as shown in Fig. 11. From Table 2, the values of n (a diffusional exponent indicating kinetic release mechanism) for P3M1 and P5M1 gels in both the stages are less than 0.50, which is an indication of Fickian diffusion. However, for P7M1 gel in stage 1, the value of n is 0.968, indicating a non-Fickian anomalous transport of drug molecules. The classification of type of transport of drug molecules from the hydrogel to the buffer medium has been taken considering the planar structure of hydrogel layer (Gami et al. 2020). For stage 2, lower value of n is observed indicating Fickian diffusion (Siepmann and Peppas 2011). So the values of n from the results suggest that mostly Fickian type of diffusion took place in the release mechanism of MCC-PVA based chemically crosslinked gels. The swelling of gel is also one of the factors for releasing the drug from the hydrogel as the results show that P5M1 gel is having higher cumulative release as well as *SR* when compared to the others. So the release mechanism of 5-Fu from hydrogel can be attributed to both controlled Fickian diffusion and swelling of gel.

Comparison of in vitro release of 5-Fu with literature

The in vitro release of 5-Fu in PBS buffer is compared with previously reported methacrylic acid (MAA) based hydrogels, and PVA based hydrogels. The

Table 2 Values of diffusion exponent (n) and correlation coefficient (R^2) of different models

Model	Parameter	P3M1		P5M1		P7M1	
		Stage 1	Stage 2	Stage 1	Stage 2	Stage 1	Stage 2
Peppas	R^2	0.985	0.984	0.991	0.993	0.998	0.950
	n	0.428	0.125	0.328	0.105	0.968	0.207
Higuchi	R^2	0.996	0.972	0.979	0.998	0.915	0.961
Zero-order	R^2	0.868	0.954	0.770	1.000	0.999	0.978
First-order	R^2	0.897	0.956	0.848	0.990	0.995	0.975

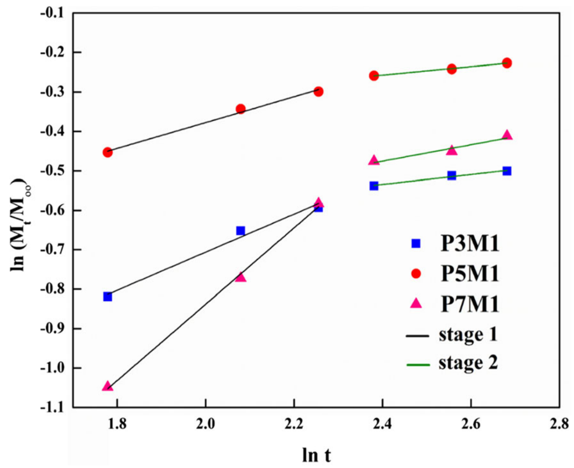


Fig. 11 Plot for the evaluation of the diffusion constant n of 5-Fu relese from hydrogel by Peppas model in two different stages

comparison is reported in Table 3. Wentao Kan and Xin Li reported less than 35% release of 5-Fu in phosphate buffer of pH = 6.86 at 25 °C within 5 h from both imprinted and non-imprinted methacrylic acid and 2-hydroxyethyl methacrylate crosslinked hydrogels (Kan and Li 2013). Minhas et al. reported the in vitro release of 5-Fu in PBS (pH = 7.4) at 37 °C from PVA and ploy methacrylic acid crosslinked hydrogels. The cumulative release of drug is around 50% in 8 h and ~ 80% release within 48 h (Minhas et al. 2013). The cumulative release of 5-Fu from P5M1 hydrogel in PBS is higher than that of the reported literature within 8 h. Therefore P5M1 hydrogel is capable of delivering more drug within a given stipulated time.

Conclusion

A series of physically and chemically crosslinked MCC-PVA hydrogels were synthesized in DMSO-NaOH/urea and water- NaOH/urea as solvents. Physical crosslinking gels synthesized by DMSO-NaOH/urea solvent did not produce promising rheological behavior whereas chemically crosslinked gels synthesized in water-NaOH/urea solvent was found to have good rheological properties. The results of rheology analysis revealed that MCC has been successfully crosslinked with PVA by showing a dominant storage modulus over the loss modulus up to the crossover point. Further, FTIR analysis also indicated that

Table 3 Comparison of 5-Fu release with literature

Author	Monomers	Drug loading concentration	Release of drug	Reference
Wentao Kan and Xin Li	Methacrylic acid and 2-hydroxyethyl methacrylate	10–50 mg L^{-1}	Less than 35% of drug within 5 h in phosphate buffer pH = 6.86 at 25 °C	Kan and Li (2013)
Minhas et al.	Polyvinyl alcohol and polymethacrylic acid	10 mg mL^{-1}	About 50% of cumulative release in 8 h and ~ 80% release after 48 h in PBS at 37 °C	Minhas et al. (2013)
Present study	Microcrystalline cellulose and polyvinyl alcohol	0.5 mg mL^{-1}	~ 60% of release within 8 h in PBS at 37 °C	–

crosslinking occurred between MCC and PVA. Morphological analysis revealed a porous structure which is responsible for the encapsulation of drug and water molecule. The results of swelling studies were complementary with rheological results, proving the lower the degree of crosslinking, higher the swelling ratio and vice versa. P5M1 gel possessed highest swelling ratio both in DI water and PBS. The loading of 5-Fu was recorded to be highest in P5M1 gels with 97.19%. Further, the in vitro release of 5-Fu, performed in PBS buffer, showed the highest cumulative release of 59.13% in 8 h. The in vitro release was influenced by the size of the drug, pore size of hydrogel and crosslinking between MCC and PVA. Further, from swelling studies, it can be said that the swelling of the hydrogel also influenced the release of 5-Fu. The kinetics of release mechanism was mostly due to the controlled Fickian diffusion.

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